Ferroelectric Digital In-Memory Computing for Scalable, Reliable, and Efficient Similarity Computation

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***Abstract*—Classification-based learning in deep neural networks, particularly few-shot learning, demands efficient similarity metrics such as Hamming distance. Conventional architectures suffer from high energy overheads due to frequent data movement between memory and processing units, hindering scalability. In-memory computing addresses this by integrating computation within memory, yet analog-based systems rely on power-hungry analog-to-digital converters (ADCs) and face scalability challenges due to device variability, especially in emerging memories. This work presents a fully digital Ferroelectric FET (FeFET)-based Logic-in-Memory (LiM) XOR cell, designed using GlobalFoundries' 28 nm technology, eliminating ADCs and ensuring robust, energy-efficient, and scalable operation. Our 2T FeFET XOR cell, applied to 4096-bit Hamming distance calculations, achieves 23× lower energy, 3× faster latency, and 14× area reduction over state-of-the-art designs. Delivering 2337 Gsamples/(s·W·mm^2) — a 300× improvement — this architecture offers a compelling solution for energy-efficient, reliable, and scalable AI hardware, driving sustainable computing.**

***Index Terms*—Complementary FeFETs, logic-in-memory, CAM, hamming distance, few-shot learning, XOR.**

I. Introduction

DEEP Neural Networks (DNNs) power various AI applications [1], [2], [3], but their performance is contingent upon large training datasets. When encountering unseen data, retraining is often required, leading to significant computational overhead and energy consumption due to frequent data transfers between memory and processing units [4], [5]. While hardware accelerators with power-hungry on-chip memory can partially mitigate latency, they remain inherently inefficient in terms of energy consumption and scalability.

In contrast, classification-based few-shot learning excels at learning efficiently from prior knowledge, reducing the need for extensive datasets and retraining [6], [7]. Classification-based learning aims to categorize input data into predefined classes [8] (Fig. 1). This process involves retrieving stored class vectors and computing the Hamming distance (HD) with a new query vector to determine the closest match [9]. HD measures the bitwise difference between vectors, providing a similarity score between the query and stored classes. The smaller the HD, the greater the similarity between the input and the stored class vector. Unlike DNNs, this approach eliminates the need for large datasets and iterative training cycles. However, conventional von Neumann architectures often limit the performance of systems relying on HD calculations, such as hyper-dimensional computing (HDC), due to memory access bottlenecks when handling large class vectors [10].

Traditional von Neumann architectures suffer from high energy consumption and latency, primarily due to extensive data transfers between memory (e.g., DRAM) and processing units required for HD calculations [9], [10] (Fig. 1). Emerging in-memory computing architectures present promising efficient solutions. By utilizing content-addressable memory (CAM) arrays, HD computation can be performed directly within the memory unit where class vectors are stored [11], [12]. The CAM output reflects the HD between the input and the stored vector based on the number of occurring matches or mismatches [9], [10]. This approach significantly reduces data movement, as only the query vector needs to be accessed or provided, leading to substantial reductions in energy consumption and latency compared to traditional von Neumann architectures.

A CAM cell implementation based on classical static random access memory (SRAM) and complementary metal-oxide-semiconductor (CMOS) technology typically requires 10 transistors [13]. This approach is volatile, bulky, and power-intensive. In contrast, emerging non-volatile memories (NVMs) such as resistive RAM (RRAM) [14], [15], [16], spin-transfer torque magnetic RAM (STT-MRAM) [17], [18], [19], and ferroelectric field-effect transistors (FeFETs) [20], [21],

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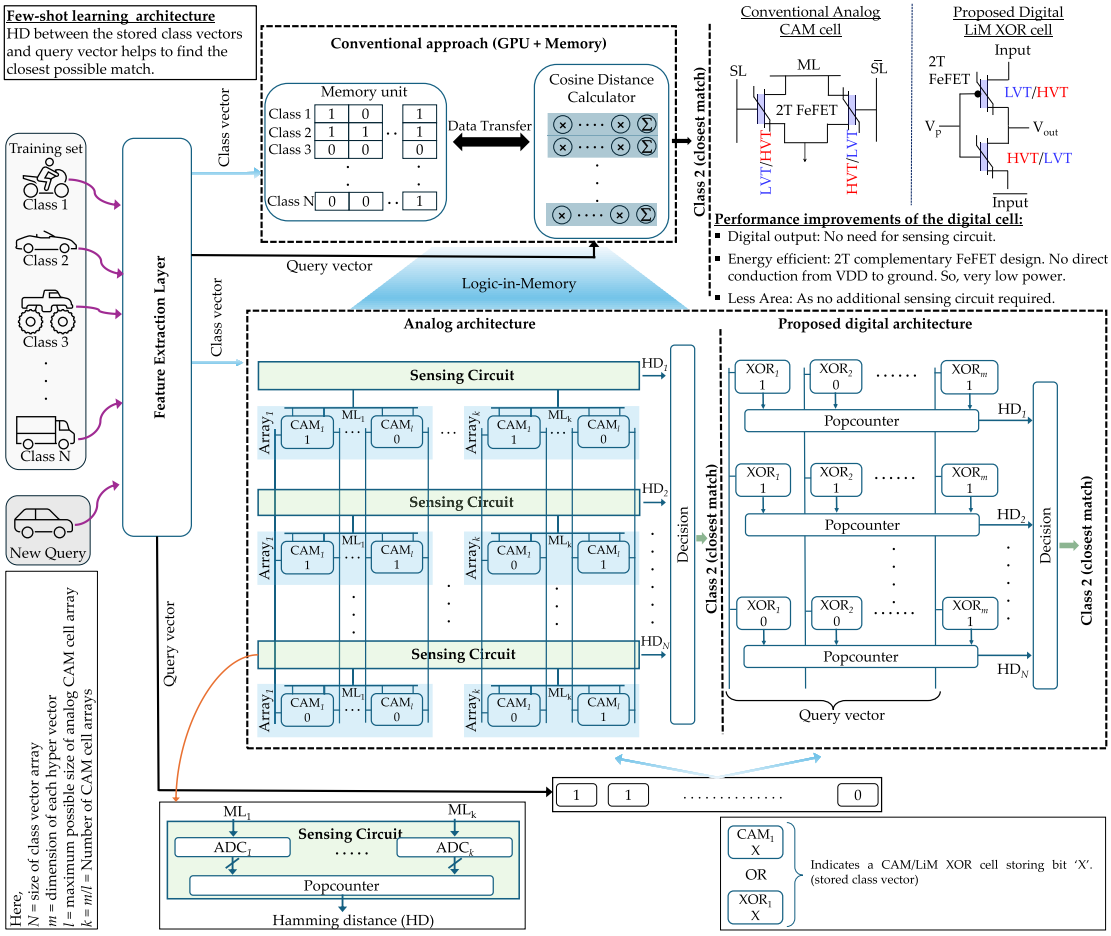


Fig. 1. Overview of the few-shot learning process. Conventional few-shot learning requires external memory and processing unit like CPU or GPU. The memory stores features (e.g., types of vehicles) extracted during training. Upon receiving a new query, the processing unit retrieves stored features from memory, compares them using cosine distance calculation, and assigns the query to the closest class. In this conventional setup, the processing unit accesses the memory for each entry to compare, resulting in significant read energy consumption and latency, which largely limits the system's scalability. CAM-based architectures store memory entries within CAM cells, enabling parallel search operations and Hamming distance (HD) computation in a single search. They are implemented using emerging NVM devices like FeFET. However, FeFET-based analog CAM cells require additional sensing circuits (i.e., ADCs) to convert match-line (ML) output to digital domains, introducing area, latency, and energy overheads. Our proposed digital LiM-based XOR cell computes HD digitally without the need for any additional sensing circuitry. Therefore, our architecture is highly efficient in terms of area, latency, and energy, and is resilient against variability, contrary to analog-based computing.

[22], [23] offers more compact and energy-efficient alternatives. However, despite their advantages, NVMs face reliability challenges and introduce system variability compared to the mature CMOS technology [24], [25], [26]. Fortunately, the inherent error tolerance [8], [27] of large class vectors in HDC applications makes NVMs a viable option.

The analog FeFET-based CAM cell utilizes two FeFETs programmed in a complementary fashion [10]. While these FeFET-based CAM cells offer advantages in terms of size and power consumption, their analog operation introduces several limitations. These designs rely on CAM cell arrays that share a single "match-line" (ML). After the query vector is applied, the ML voltage will drop according to the number of positions that query is different from the stored class bits. Then, HD is calculated by sensing the voltage on this ML [10]. However, accurately converting this analog voltage to a digital signal representing the HD is challenging, as it requires power/area-hungry ADCs. In larger arrays, the ADC struggles to distinguish between subtle variations in the ML voltage drop, particularly when compounded by variability in FeFET devices [9]. As the size of the CAM array increases, the ML voltage starts to drop non-linearly until reaching a lower threshold, beyond which it becomes insensitive to additional discharges. This saturation effect forces designers to use smaller arrays, undermining the scalability potential of large-scale HDC learning systems. Moreover, ADC-based sensing circuits introduce significant overhead in terms of both area and energy consumption. In a CAM array comprising six cells, the accompanying ADC occupies an area of 357 μm^2, while the CAM cells themselves use only 26 μm^2 (refer section: IV-B). This highlights the fundamental challenge of analog computing when paired with emerging NVM devices, underscoring the critical need to replace conventional analog CAM-based HD calculators with digital Logic-in-Memory (LiM) alternatives [28], [29], [30], [31], [32], [33]. Digital LiM architectures not only enhance energy and area efficiency by eliminating the need for sensing circuits but also offer significantly higher reliability, leveraging the robustness of digital computing.

Our Key Contribution: In this work, we propose a FeFET-based digital LiM XOR cell to overcome the limitations of analog CAMs. The digital XOR gate directly outputs a

0 (match) or 1 (mismatch) based on the comparison with the input query vector, eliminating the need for power-hungry ADCs and improving resilience to variations in FeFET characteristics. By removing the need for additional sensing circuits, the design significantly reduces both silicon footprint and energy consumption, delivering substantial efficiency gains for in-memory computing applications. The fully digital operation of the cell enhances its robustness to process variations, ensuring more reliable computational performance. *We utilize complementary FeFETs (pFeFET and nFeFET) to implement a compact digital XOR gate using only two FeFETs*. While classical CMOS-based XOR requires a significant number of transistors, our FeFET-based LiM provides a highly efficient alternative. This optimized design not only reduces area but also significantly lowers energy consumption, making it well-suited for large-scale classification-based learning systems.

In summary, this work presents the design of a fully digital HD computation unit using a novel FeFET-based digital XOR gate, which forms the building block of large-size array, leveraging GlobalFoundries' 28 nm FeFET technology. We conduct a comprehensive analysis of the cell-level performance metrics of the proposed digital LiM-based XOR gate and compare it with traditional analog CAM-based designs. Furthermore, simulations with 4096-bit class vectors provide estimates of system-wide latency, area, and energy consumption during HD calculations. Finally, the performance of our design is comprehensively benchmarked against other contemporary LiM architectures, demonstrating its competitive advantages in efficiency and scalability.

The contributions of this work cover the following levels:

- **Device Characterization:** We have characterized both p-type and n-type FeFETs fabricated using GlobalFoundries' 28 nm SLPe technology. This characterization involves evaluating 200 devices from a 300 mm wafer to assess how variations affect threshold voltage (V_{TH}) and memory window performance. Additionally, we examine the influence of silicon–germanium (SiGe) channel on p-type device behavior to quantify its impact on FeFET and our LiM-based XOR design.
- **Cell-Level Evaluation:** We have designed a novel digital-based LiM XOR gate using p-type and n-type FeFETs. To capture the behavior of fabricated FeFETs, we employ a combined model that integrates the Preisach model [34] for the ferroelectric capacitor and the industry-standard BSIM compact model [35], [36] for the transistor. We then compare the latency, area, and power consumption of our design with existing analog-based CAM design, while also evaluating the impact of device variability on the cell's reliability.
- **Circuit-Level Analysis:** Using the proposed digital-based LiM XOR cell, we have constructed an array to calculate the HD for large class vectors consisting of 4096 bits. For comparison, we have also designed an analog CAM array, complete with its associated sensing circuits, to calculate HD using the traditional analog-based FeFET approach where two n-type FeFETs are used. This design considers FeFET variability and ADC sense margin optimization. Additionally, we implemented a digital popcounter circuit to compute the final HD (Fig. 1). Layout design and post-layout simulations are performed using commercial EDA tools [37] to estimate area, latency, and energy consumption for both designs.

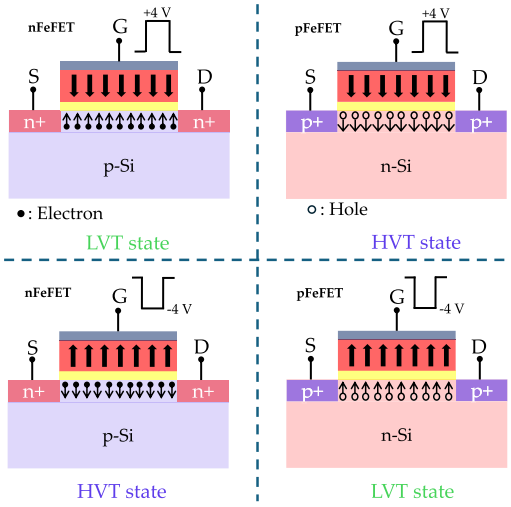


Fig. 2. Proposed complementary FeFET programming scheme. The polarity of programming pulses affects the channel conductivity differently in n-type and p-type FeFETs. A positive pulse generates a polarization field within the ferroelectric layer directed toward the transistor's channel, facilitating inversion in n-type FeFETs but opposing it in p-type FeFETs. Thus, a positive pulse programs nFeFETs to the LVT state and pFeFETs to the HVT state. Conversely, a negative pulse reverses this effect.

- **System-Level Comparison:** Finally, we conducted a comprehensive evaluation of the overall efficiency of our digital LiM-based XOR design, comparing it against state-of-the-art analog CAM designs and other contemporary LiM-based approaches. This benchmarking ensures a thorough comparison in terms of energy efficiency, scalability, reliability, and area utilization.

II. Background and Preliminaries

A. FeFET Fundamentals

FeFETs are built upon the conventional metal-oxide-semiconductor field-effect transistors (MOSFETs) by incorporating a ferroelectric layer (e.g., HfO_2) between the gate and insulating oxide (SiO_2) [21]. This ferroelectric material exhibits two polarization states, switchable by an electric field exceeding the coercive field. Applying appropriate voltage pulses to the FeFET gate modulates the ferroelectric polarization which influences the transistor channel's conductivity. In n-type FeFETs (nFeFETs), a positive pulse generates a polarization field within the ferroelectric layer pointing towards the channel of the transistor (Fig. 2). This polarization field facilitates inversion in the silicon channel and results in a low-V_{TH} (LVT) state. Conversely, a negative pulse reverses the polarization field direction, hindering the inversion and leading to a high-V_{TH} (HVT) state. The effect of the polarity of pulses is reversed for p-type FeFETs (pFeFETs) due to their hole-induced inversion mechanism. So, applying a pulse of appropriate magnitude and polarity, one can program the FeFETs to operate at either LVT or HVT state. The detailed programming scheme is shown in Fig. 2. These distinct LVT and HVT states, separated by a memory window, make FeFETs attractive for non-volatile memory cells, as they retain their programmed state even after the supply voltage is turned off. This non-volatility and low power consumption make FeFETs particularly suitable for in-memory computations.

TABLE I
FIGURES OF MERIT FOR RRAM TECHNOLOGY

Technology	90 nm [1]	65 nm [2]
Cell structure	2T2R	2T2R
Cell area	0.41 μm^2	1.1 μm^2
Word size	64-bits	128-bit
Total area (mm^2)	5.94 for 1Mb	NA
Search latency (ns)	1.9 @V_{DD}=1.2 V	0.62 @V_{DD}=0.7 V

B. CAM as an HD Calculator

Content-based addressing is a key function of memory modules, enabling efficient searches by comparing an input vector against stored vectors to identify the closest match [38]. Unlike conventional architectures that require external data transfers, CAMs perform in-memory searches using binary comparators (XOR), significantly reducing data transfer overhead between memory and processing units. In a CAM array, cells within a row share a ML, which is sensed by an ADC (Fig. 1). During a search, MLs are precharged to a high voltage, and search lines ($SL/\overline{SL}$) are activated based on the input query. The ML voltage discharges to a specific level depending on the number of bit-wise matches or mismatches between the query vector and the stored class vector. The comparisons between stored data and the query are performed through bitwise XOR operations within the memory, effectively computing the HD.

Conventional CAMs based on CMOS technology offer high-speed search capabilities but suffer from substantial area overhead and power consumption [13]. To overcome these limitations, NVM-based CAMs (NV-CAMs) have emerged as a promising alternative, providing compact designs, enhanced energy efficiency, and non-volatile data retention. NV-CAMs utilize various NVM technologies, including RRAM [14], [15], [16], STT-MRAM [17], [18], [19], and FeFETs [10], [20], [21], [22], [23]. These advanced memory technologies offer improved performance, making NV-CAMs an attractive solution for high-density and energy-efficient applications. The 2T-2R configuration is a widely used structure for CAM cells, incorporating two select transistors and two resistive storage elements, typically based on RRAM [39] or STT-MRAM [19]. These resistive storage-based CAM designs are fully compatible with standard CMOS processes and occupy less area compared to conventional CMOS-based CAMs. For instance, the work presented in [40] reports a 2T2R RRAM-based TCAM fabricated in a 90 nm CMOS process. The design achieves a cell area of 0.41 μm^2 and, for a 1 Mb array, occupies a total silicon area of 5.94 mm^2. It demonstrates a search latency of 1.9 ns at a supply voltage of 1.2 V. Similarly, the study in [41] presents a 2T2R-based TCAM implemented in 65 nm technology, which, for a 128-bit word length, achieves a search latency of 0.62 ns at 0.7 V and an area of 13.7 μm^2 per matched line. The performance metrices of the RRAM technology are tabulated in Table I. Similarly, the 2T FeFET-based analog CAM cell also represents a promising alternative, offering both high area efficiency and reduced energy consumption [10], [20], [21], [22], [23]. However, despite these benefits, several challenges persist. CAM-based crossbar arrays inherently require additional peripheral sensing circuitry—specifically, sense amplifiers and ADCs—to detect the currents/voltages on the MLs and convert them into corresponding digital outputs. These peripheral circuits are not only bulky but also prone to variability, posing significant challenges for designers seeking to develop efficient, accurate, and scalable NVM-based HD computing architectures.

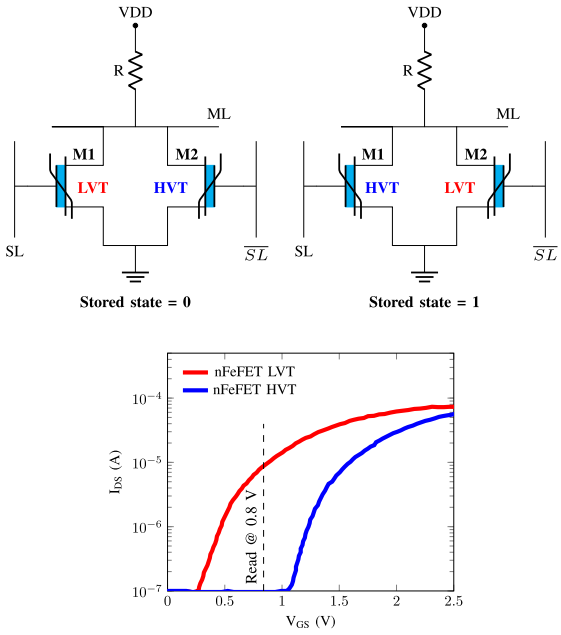


Fig. 3. The working principle of the existing analog CAM cell. The CAM cells consist of two nFeFETs programmed in complementary states. To store a logical *0, M1/M2 are programmed in LVT/HVT states, respectively, while to store a logical *1, M1/M2 are programmed in HVT/LVT states, respectively. The I_{DS}-V_{GS} characteristics illustrate the LVT and HVT responses of the transistors. Read operations are performed by applying appropriate read voltages (0 V or 0.8 V) at the gates of the transistors. When a logical *0 is stored, the ML discharges through M1 during a mismatch (SL = 0.8 V). Similarly, when a logical *1 is stored, the ML discharges through M2 during a mismatch (SL = 0 V).

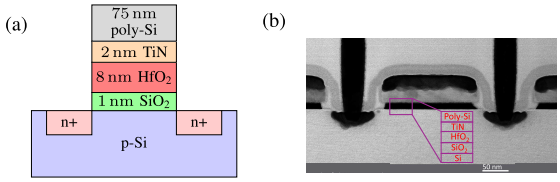
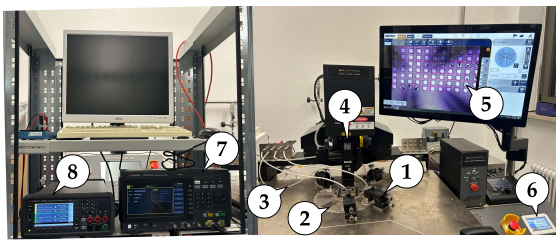


Fig. 4. (a) Material stack of the nFeFET, (b) TEM image of our fabricated FeFET in 28 nm technology.

C. Existing State-of-the-Art FeFET-Based Analog CAM

FeFET is a promising candidate for CAMs owing to its non-volatility, CMOS compatibility, and excellent I_{ON}/I_{OFF} ratio [21]. The conventional FeFET-based CAM cell comprises two n-type FeFETs programmed in complementary states [10], as illustrated in Fig. 3. When transistors M1 and M2 are programmed to LVT and HVT states, respectively, the CAM stores a logical *0. Conversely, the CAM stores a logical *1 when transistors M1 and M2 are in HVT and LVT states, respectively. The drains of both transistors are connected to VDD through a resistor. During the read operation, a logical *0 (0 V) input at the SL line of transistor M1 will keep the transistor in the OFF state, as demonstrated by the transfer characteristics in Fig. 3. Although $\overline{SL}$ is at 0.8 V, transistor M2 remains OFF due to its HVT state. Therefore, when a matching read input is provided to the CAM cell, both transistors are OFF, and the ML voltage remains constant. In contrast, if a



1. Probe arm
2. The FeFET wafer loaded inside
3. Triaxial cables
4. Microscope
5. Contact pads of the DUTs
6. Temperature monitor
7. Keysight EDU33212A waveform generator
8. Keysight B2902B SMU

Fig. 5. Measurement setup for FeFET characterization. An MPI TS3000-SE 300 mm probe station is used for wafer probe measurement of the FeFET devices. The necessary biases are provided using Keysight waveform generator and SMU. Triaxial cables are connected to the probe arms to transfer the signal to the DUT contact pads. The 300 mm wafer containing the DUTs is loaded on the thermal chuck, residing inside the probe station.

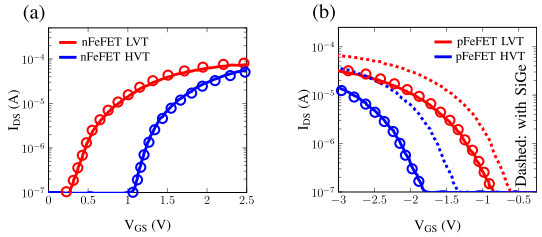


Fig. 6. Transfer characteristics of the measured FeFET devices. The I_{DS}-V_{GS} characteristics of GlobalFoundries' 28 nm technology FeFETs: (a) nFeFET device with dimensions 500 nm x 500 nm, and (b) pFeFET with dimensions 500 nm x 500 nm. Symbols represent the compact model simulation, and solid lines represent the measured data. Dashed lines represent characteristics corresponding to SiGe channels, which reduce the transistor's threshold voltage. Pulses of magnitude $\pm$ 4 V are used for transistor programming.

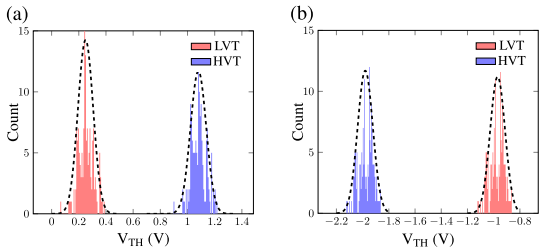


Fig. 7. Experimentally measured device-to-device variation of FeFET devices. (a) Distribution of LVT and HVT for nFeFET. (b) Distribution of LVT and HVT for pFeFET. The distribution across 200 devices exhibits a standard deviation of approximately 57 mV.

read voltage of 0.8 V (logical *1) is applied to SL, transistor M1 will conduct, discharging the ML voltage to a lower level. Similarly, when a logical *1 is stored, M1 is programmed to the HVT state, and M2 is in the LVT state. A logical *0 input at SL will keep M1 OFF; however, the 0.8 V input at $\overline{SL}$ will turn M2 ON, which is in the LVT state. Thus, a mismatch will also discharge the ML voltage in this scenario. Therefore, the ML line discharges only when there is a mismatch between the stored bit and the read input; otherwise, the ML maintains a stable voltage close to VDD.

In the design of a CAM array, the ML is shared among a group of CAM cells, as depicted in Fig. 14. Consequently, the ML voltage settles depending on the number of mismatches, causing a corresponding number of CAM cells within the array to discharge. Following this, an ADC detects the ML voltage (Fig. 14), determining the number of matched or mismatched input bits. The operation of the CAM cell is inherently analog, with the ADC responsible for sensing the voltage on the ML and converting it into a digital signal. This analog operation presents significant challenges when multiple CAM cells share a single ML. As the number of CAM cells sharing the ML increases, the discharge current eventually becomes so large that the voltage drop across the resistance R becomes very large. When (VDD - $I_{discharge} \times$ R) becomes small enough, the ML voltage saturates, rendering it insensitive to further discharges from additional CAM cells. This phenomenon limits the scalability of the CAM array. Additionally, the variability of FeFETs complicates the ADC's ability to accurately distinguish between adjacent states, raising serious reliability concerns (detailed in Sec. IV-B).

III. Proposed Architecture and Design Methodology

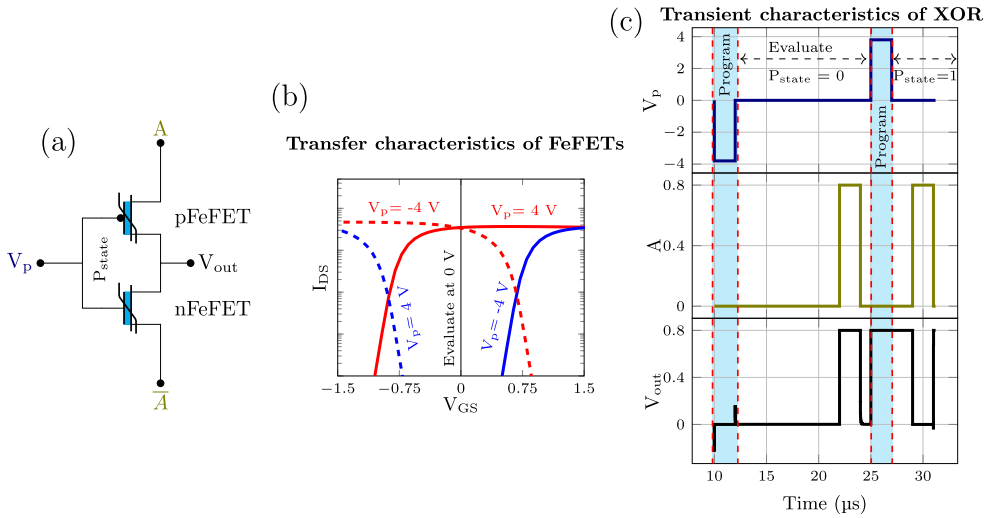
A. Structural Details of the FeFET

This work utilizes FeFETs fabricated on GlobalFoundries' 28 nm high-K/metal gate (HKMG) SLPe technology platform. The gate stack comprises a thin interlayer of SiO_2 (~ 1 nm) and a thicker layer of doped HfO_2 (~ 8 nm) as the ferroelectric material (Fig. 4 (a)). The gate cap consists of a combination of TiN and silicided polysilicon. Fig. 4 (b) shows the transmission electron microscopy (TEM) image of a cross-section of the FeFET integrated into the 28 nm HKMG technology.

B. Measurement Setup and Characterization

We have used an MPI TS3000-SE probe station (Fig. 5) for the electrical characterization of the fabricated FeFET devices. The 300 mm wafer, which contains the DUTs, is securely mounted on the probe station's thermal chuck. During the entire measurement process, the thermal chuck is precisely maintained at a temperature of 300 K using a dedicated temperature controller. Electrical connections to the DUTs are established via MPI MP40 probe arms, which facilitate contact between the probe tips and the device's contact pads. Four probe arms are employed to simultaneously access the drain, gate, source, and substrate terminals of the DUTs. These probe arms are connected to the signal inputs using triaxial cables to ensure low-noise measurements. Bias voltages required for the characterization are provided by a Keysight EDU33212A waveform generator and a Keysight B2902B SMU. Both instruments are connected to a computer using a USB interface to control them using custom Python scripts. A detailed illustration of the entire measurement setup is shown in Fig. 5.

The waveform generator drove the gate, while the SMU connected to the drain measured current. Program and erase pulses were applied to the gate, followed by a gate voltage (V_{GS}) sweep to capture I_{DS}-V_{GS} characteristics. The SMU applied a drain bias and measured I_{DS} at each V_{GS}. For nFeFETs, a +4 V program pulse and −4 V erase pulse (both 100 μs wide) were used. In pFeFETs, the polarity was reversed. To minimize charge trap influence, a 200 ms delay was introduced between program/erase cycles and the V_{GS} sweep for read. During readout, V_{GS} was swept from 0 V to 2.5 V (0 V to −3 V for pFeFETs) in 50 mV steps over 50 ms. The drain voltage was held at 100 mV, and the SMU recorded drain



Truth table for complementary FeFET-based XOR

V_p [Programming]	V_{out} [Evaluation @V_p= 0 V]	FeFET states
4 V(P_{state}= 1)	$\overline{A}$	nFeFET: LVT; pFeFET: HVT
-4 V (P_{state}= 0)	A	nFeFET: HVT; pFeFET: LVT

Fig. 8. Digital LiM-based XOR gate designed using complementary FeFETs. (a) The design of the digital XOR cell (b) The I_{DS} - V_{GS} characteristics of the complementary FeFETs for different programming pulses and their alignment at the evaluation phase (V_{GS} = 0 V.) (c) The output of the XOR gate based on different activation inputs corresponding to various programmed states. The table shows the states of the FeFETs during different programming states and the corresponding output at V_{out}.

current to generate I_{DS}-V_{GS} characteristics. The I_{DS} - V_{GS} characteristics of 500 nm x 500 nm nFeFET and pFeFET devices are depicted in Fig. 6 (a) and (b), respectively. As shown in Fig. 6 (b), the SiGe channels reduce the overall V_{TH} of the transistor, providing further control over V_{TH} tuning beyond the work function of the metal gate. We investigated the variability of the fabricated FeFETs by measuring 200 devices. The measured LVT and HVT distributions are presented in Fig. 7. The standard deviation of the V_{TH} values across the devices is approximately 57 mV. This analysis offers insights into device-to-device variations, which are critical considerations when designing FeFET-based circuits.

C. FeFET-Based Digital Logic-in-Memory XOR

As discussed earlier, FeFETs exhibit two distinct programmable V_{TH} states, depending on the applied programming voltage. Applying a specific input voltage to a programmed FeFET allows it to perform certain Boolean logic operations based on its stored memory state. In the context of few-shot learning and HDC algorithms, calculating HD between a stored class vector and an input query vector requires a logical XOR operation.

To achieve that, we leverage complementary FeFETs (p-type and n-type) to design efficient digital-based LiM XOR gates tailored for HD calculations. This LiM-based architecture provides advantages in terms of area and power consumption when compared to traditional analog-based CAM. Fig. 8 illustrates a simple digital XOR gate design using only two FeFETs. These FeFETs are pre-programmed using appropriate voltage pulses (V_p) applied to their gates. During operation, the activation input (A) and its complement ($\overline{A}$) are applied to the drain terminal of the pFeFET and the source terminal of the nFeFET, respectively. The XOR operation is realized through two distinct phases:

- **Program:** Initially, the FeFETs are programmed by applying ± 4 V. Applying V_p = +4 V configures the system to P_{state} = 1, while V_p = −4 V configures it to P_{state} = 0. The corresponding FeFET polarization states are tabulated in Fig. 8.
- **Evaluation:** Once programmed, the evaluation is performed at V_p = 0 V. Depending on the programmed state, either the pFeFET or the nFeFET will turn ON during the evaluation phase and pass the corresponding input (A or $\overline{A}$) to the output.

The transfer characteristics plot of the FeFETs in Fig. 8 (b) shows the I_{DS}-V_{GS} configurations under different programming voltages V_p. For example, programming the FeFETs by applying V_p = +4 V configures the nFeFET in the LVT mode and the pFeFET in the HVT mode. During the evaluation phase with V_p = 0 V, the nFeFET will be ON (as shown in the transfer plot) and will pass $*\overline{A}$ to the output. In this state, the pFeFET remains OFF, preventing *A from reaching the output. When programmed with V_p = −4 V, the roles are reversed, allowing *A to pass to the output. The output (V_{out}) corresponding to different V_p values and associated FeFET states are tabulated in Fig. 8. Consequently, the Boolean expression for V_{out} is $V_{out} = P_{state} \cdot \overline{A} + \overline{P_{state}} \cdot A = A \oplus P_{state}$. The waveforms in Fig. 8 illustrate the functionality of the XOR gate, clearly showing the program and evaluation stages. The complementary FeFET-based XOR uses only four transistors (two FeFETs and two conventional CMOS transistors for the inverter operation), thereby saving area, reducing power consumption, and improving speed.

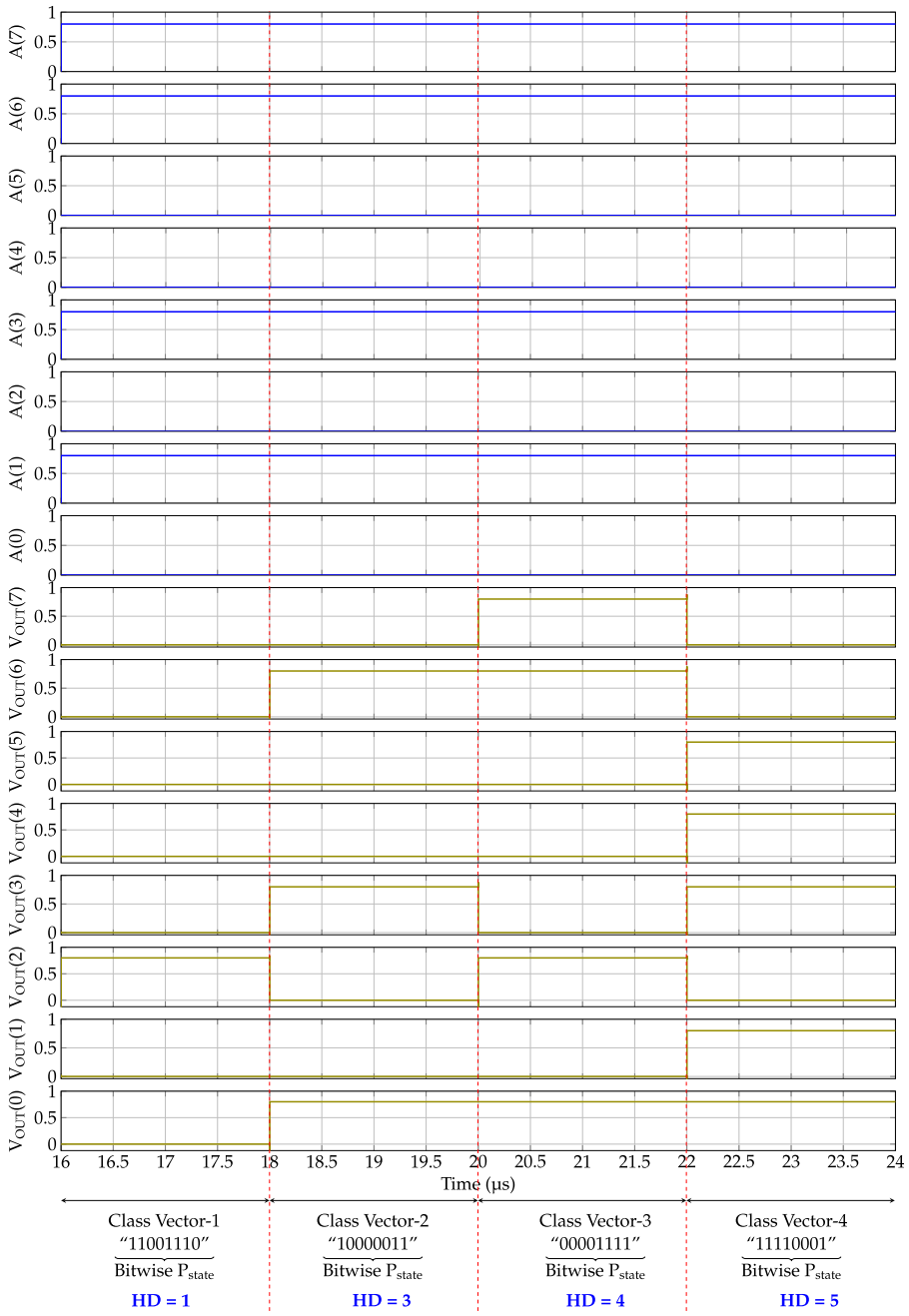


Fig. 9. HD calculation using the proposed LiM-based architecture for an 8-bit class vector. Four representative class vectors are considered. The input query vector, denoted as A(7) to A(0), is applied to the system, and the corresponding bitwise XOR outputs, $V_{OUT}(7)$ to $V_{OUT}(0)$, are shown for each class vector across distinct time intervals.

D. HD Calculation Using Our Digital-Based LiM XOR

The HD between two binary vectors represents the number of mismatching bits. It is calculated using a bit-wise XOR operation, where matching bits yield a logic *0, and mismatches result in a logic *1. Therefore, a higher number of *1s in the output indicates a greater degree of dissimilarity between the two vectors. In this work, we utilize our digital-based LiM XOR cells for HD computation. Pre-programmed class vectors are stored within these cells, and query vectors are fed as inputs to generate the digital output (Fig. 1). The accuracy of HDC algorithm improves with higher vector dimensions. Here, we consider a 4096-bit class vector, with each LiM XOR cell performing a bit-wise XOR operation. Consequently, 4096 such cells are required for the full HD calculation, resulting in a 4096-bit output array. To determine the HD, we count the number of *1s in the output array. For this purpose, we introduce a 4096-bit digital popcounter circuit.

The circuit functionality of the LiM-based digital HD calculation is implemented using SPICE-based simulations. The FeFET experimental characteristics are emulated with a calibrated compact model. We have utilized a Preisach model

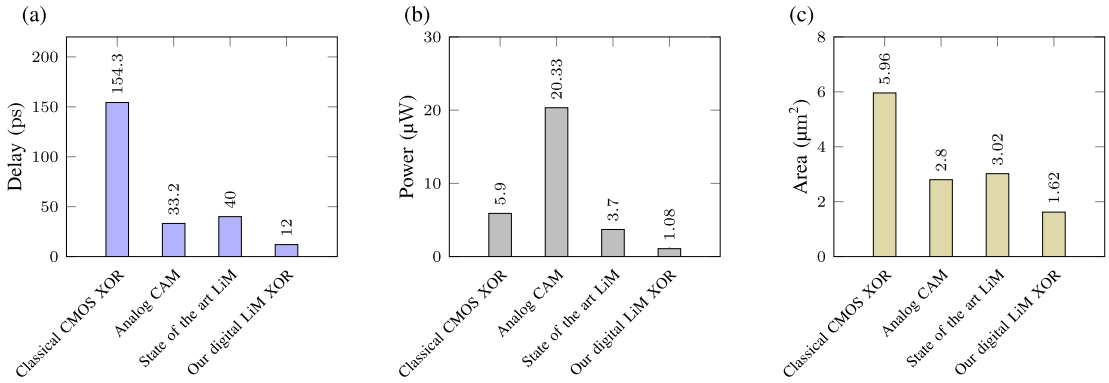


Fig. 10. Cell-level figure of merit comparison. Our proposed XOR cell achieves reductions in delay, power dissipation, and area by approximately 13x, 5.5x, and 3.7x, respectively, compared to the CMOS-based XOR. Similarly, compared to the analog CAM cell [10], our design shows improvements of 2.8x, 19x, and 1.7x in latency, power, and area, respectively. The higher power dissipation in the analog CAM is due to continuous current flow from the ML to the ground during read operations. Our design also demonstrates similar improvements compared to the state-of-the-art digital LiM-based XOR [33].

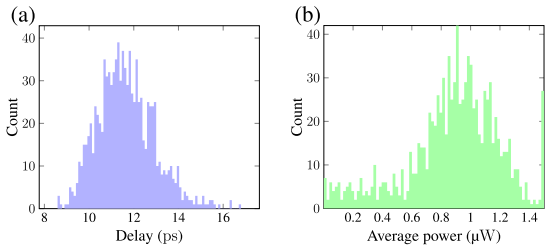


Fig. 11. Impact of variability on the designed XOR cell performance. (a) Variation in cell delay. (b) Variation in average power dissipation. Results are derived from SPICE-based Monte Carlo simulations with 1000 samples, introducing a standard deviation of 57 mV in V_{TH}, consistent with experimental observations from measurements.

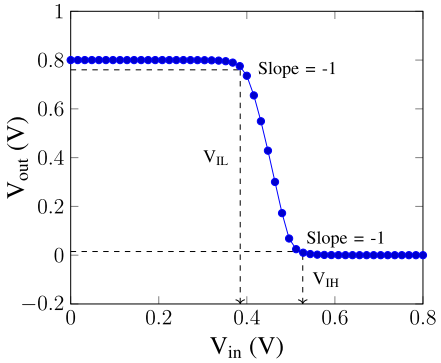


Fig. 12. Voltage transfer characteristics (VTC) of the proposed LiM XOR cell. The query input *A (V_{in}) is swept from 0 V to 0.8 V. The V_{IL} and V_{IH} are 0.385 V and 0.527 V, respectively. Based on these values, the calculated noise margins are NML = 0.385 V and NMH = 0.273 V, respectively.

[34] for the ferroelectric capacitor in the FeFET and coupled it with the 28 nm PDK to represent the underlying transistor. The modeling methodology is detailed in the Appendix. The model parameters are adjusted to accurately replicate the measured FeFET characteristics, and the gate work function is tuned to achieve the desired functionality. This calibrated model is then used in SPICE simulations to test the proposed design and extract reports on latency, area, and power dissipation. Device-to-device variability is incorporated by introducing variations in the threshold voltage of the transistors, with a standard deviation matching the measured data. Additionally, we have implemented and simulated a classical CMOS-based XOR gate and an analog-based CAM cell using the same FeFET calibrated model to ensure fair comparisons against our proposed design. The area reports for these designs are obtained by laying out the designs, using the GlobalFoundries' 28 nm process design kit (PDK). For the digital popcounter circuit, we have written the corresponding register-transfer level (RTL) in Verilog and synthesized it using the commercial logic synthesis tool [42], which provides the area, latency, and power dissipation required for calculating the final HD.

To enhance the understanding and validate our proposed LiM-based HD calculator architecture, we have performed a simulation using representative class vectors encoded in 8-bit format. As shown in Fig. 9, we consider four class vectors, each 8 bits long:

- Class Vector-1 ("11001110")
- Class Vector-2 ("10000011")
- Class Vector-3 ("00001111")
- Class Vector-4 ("11110001")

Each bit in these class vectors is mapped to the corresponding FeFET programming state, as described in section-III (C) of the manuscript. Specifically, a programming state $P_{state} = 1$ corresponds to a bit value of 1, while $P_{state} = 0$ corresponds to a bit value of 0. During the programming phase, all four class vectors are bitwise programmed into their respective FeFET-based LiM cells. In the subsequent evaluation phase, an 8-bit query vector A=11001010 is applied. Each LiM cell performs a bitwise XOR operation between the query vector and the stored class vector. Since each class vector is 8 bits wide, 8 LiM cells are used per vector to perform the XOR operation in parallel.

The 8-bit output of this XOR operation is denoted as $V_{OUT}(7)$ to $V_{OUT}(0)$, indicating the bitwise result for each position. These outputs are shown in Fig. 9 across time segments as follows:

- 16 μs–18 μs: Output for Class Vector-1
- 18 μs –20 μs: Output for Class Vector-2
- 20 μs –22 μs: Output for Class Vector-3
- 22 μs –24 μs: Output for Class Vector-4

From the simulation results, we observe that the XOR output with Class Vector-1 has the minimum number of ones (= 1), indicating a HD of 1. Hence, Class Vector-1 is identified as the closest match to the input query vector.

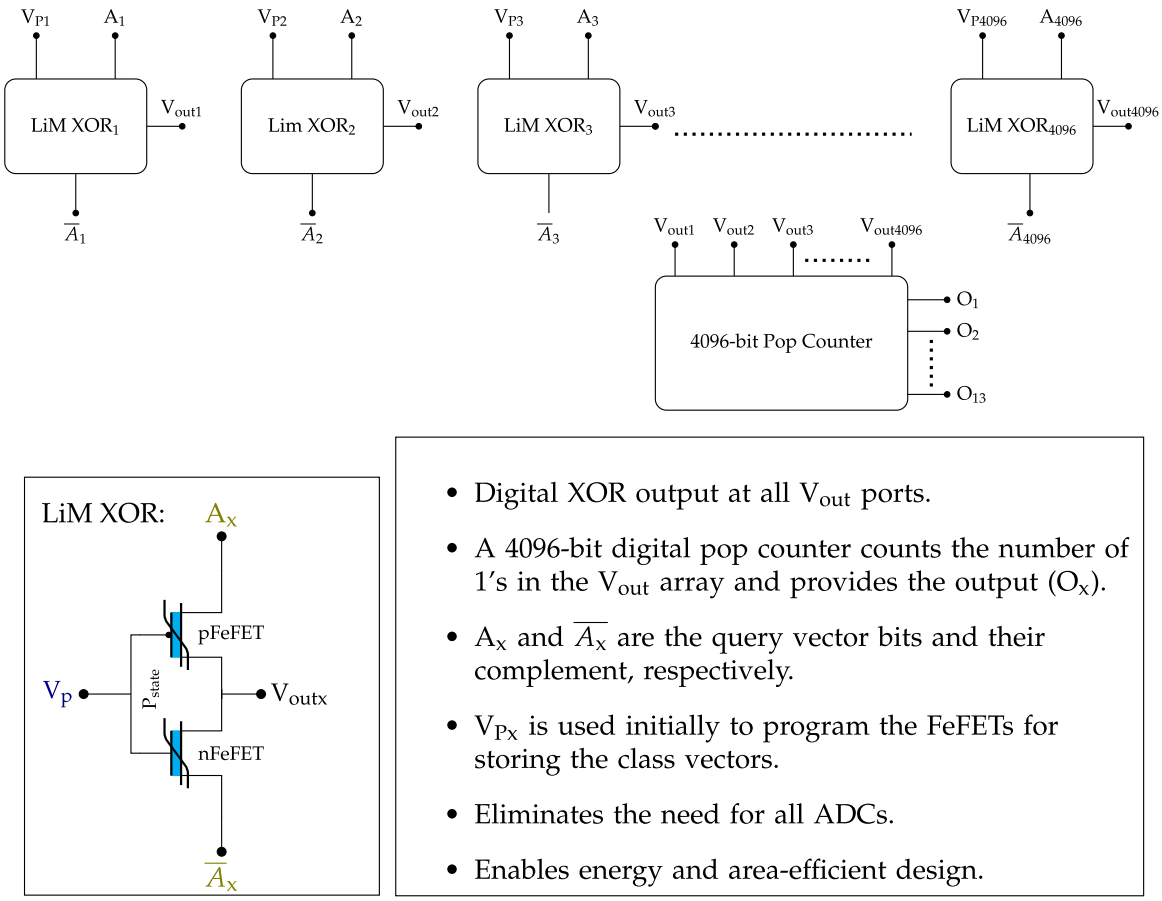


Fig. 13. Array architecture for HD computation using the proposed LiM XOR cells. The class vector is represented by 4096 pre-programmed LiM cells via ports V_{P1} to V_{P4096}. The query vector is applied through inputs A_1 to A_{4096}, and the bitwise XOR results are obtained at outputs V_{out1} to $V_{out4096}$. A 4096-bit digital pop counter counts the number of 1's in the V_{out} array and provides the final output (O_x).

IV. Performance Evaluation

A. Cell Level

To assess system-level performance improvements, it is essential to conduct a comparative evaluation between the proposed design and the existing ones. Our digital-based LiM XOR utilizes two FeFETs (1 pFeFET and 1 nFeFET) along with one CMOS inverter to invert the query vector input. Calibrated SPICE simulations indicate that our designed cell exhibits a delay of 12 ps (Fig. 10 (a)). Furthermore, our complementary FeFET-based design improves energy efficiency. Unlike the analog-based CAM cell, our digital-based LiM XOR cell does not establish a direct conduction path from the supply to the ground during read operations, as one of the FeFETs remains non-conductive at all times. The average power dissipation of our design is 1.08 μW, which is attributed mainly to dynamic power. In contrast, the analog CAM cell consumes significantly more power during read operations because the ML discharges through the FeFET(s) during a mismatch, resulting in an average power dissipation of approximately 20 μW—about 19x higher than that of our digital-based LiM XOR cell (Fig. 10 (b)). The CAM cell also shows a delay of 33.2 ps, approximately 2.7 times greater than our digital-based LiM XOR cell. Additionally, simulations of an existing LiM-based 6T dynamic XOR gate, as proposed in [33], show that the 6T XOR cell exhibits a delay of 40 ps and an average power dissipation of 3.7 μW (Fig. 10). Clearly, our complementary FeFET-based digital LiM XOR outperforms other existing LiM designs in terms of speed and energy efficiency. The proposed design also demonstrates a reduced area footprint compared to other current designs, as illustrated in Fig. 10 (c) (see Fig. 16 and Fig. 17 for layout information). The impact of device-to-device variability on the output of the proposed XOR cell is presented in Fig. 11. These results were obtained through SPICE-based Monte Carlo simulations with 1000 samples, introducing a standard deviation of 57 mV in the V_{TH} of the FeFETs, as measured in our 300 mm wafer.

Noise margins of the gates are also important to evaluate the robustness of their performance. We have calculated the noise margins of the proposed LiM XOR to estimate the robustness of the cell against any noise source. Fig. 12 illustrates the voltage transfer characteristics (VTC) of the LiM cell as the query input (A) is swept from 0 V to V_{DD} = 0.8 V. The output high and low voltages are defined as V_{OH} = 0.8 V and V_{OL} = 0 V, respectively. The input low voltage (V_{IL}) is determined as the input voltage at which the slope of the VTC, $\frac{dV_{out}}{dV_{in}}$, equals -1 for the first time—marking the transition to a logic low region. Similarly, the input high voltage (V_{IH}) is identified as the point at which the same condition ($\frac{dV_{out}}{dV_{in}} = -1$) occurs for the second time, corresponding to the logic high transition. Based on these definitions, the noise margins are computed as follows:

- NML = V_{IL} - V_{OL}
- NMH = V_{OH} - V_{IH}

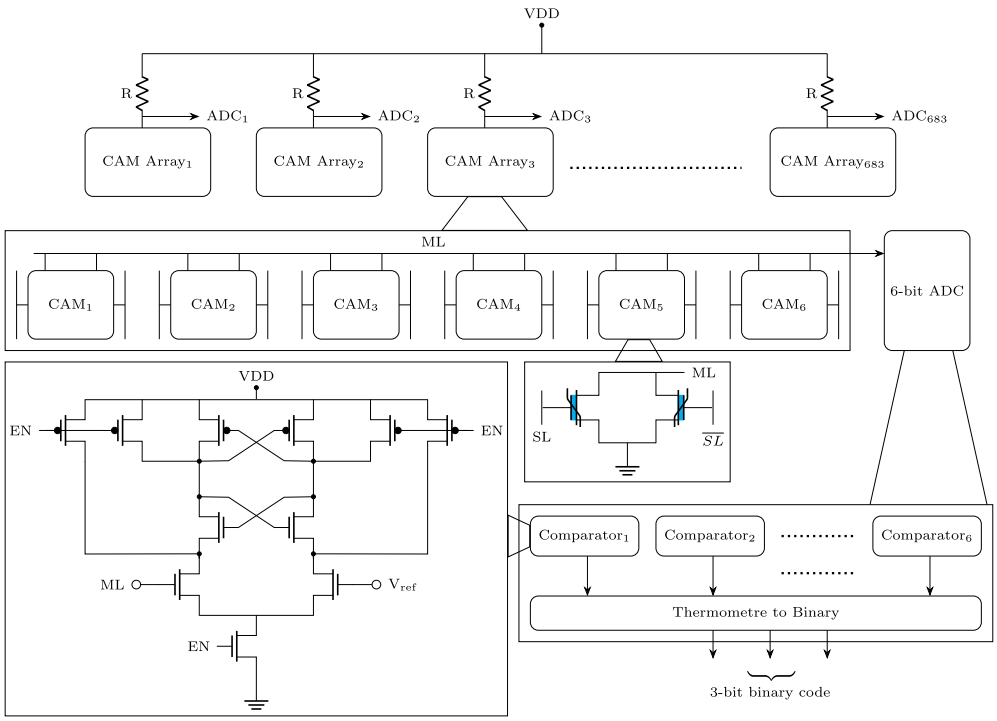


Fig. 14. Analog CAM circuit with associated sensing circuitry. The CAM array size is limited to 6 cells due to the ADC sense margin and FeFET variability in this technology. To compute the HD for a 4096-bit vector, 683 CAM arrays of size 6 are required. Each array's output is linked to a 6-bit ADC, which detects ML voltage levels corresponding to the number of mismatches. The resulting 3-bit binary code from the ADC is fed into a popcounter circuit for the final HD evaluation.

From Fig. 12, it is observed that V_{IL} = 0.385 V, and V_{IH} = 0.527 V. This results in NML = 0.385 V and NMH = 0.273 V for the proposed LiM XOR.

B. System Level

As depicted in Fig. 1, the XOR cell is the unit cell that performs the first stage in HD calculations between a stored class vector and an input query vector. The size or dimension of the vector significantly influences the inference accuracy of the HDC system, with larger vector dimensions correlating to higher inference accuracy [9]. For this study, we utilize a vector size of 4096 bits. To compute the HD between two 4096-bit vectors, an array of 4096 digital XOR cells is required. Initially, the proposed XOR cells are programmed into states (P_{state} = 0/1) that represent the class vector. The query vector and its complement are then applied to the source and drain of the n- and p-type FeFETs, respectively (Fig. 8).

Fig. 13 illustrates the array-level architecture of our proposed LiM-based system for computing the HD of a 4096-bit class vector. Each bit of the class vector is mapped to an individual LiM XOR cell, resulting in a total of 4096 LiM cells per array. These cells are pre-programmed to represent a specific class vector. During the evaluation phase, a 4096-bit query vector is applied simultaneously to the array, as shown in Fig. 13. Each LiM cell performs a bitwise XOR operation in parallel, and the output V_{out} collectively forms the result of the similarity comparison. The XOR array outputs a 4096-bit binary string, from which we determine the HD by counting the number of *1s. This counting is performed using a 4096-bit digital popcounter circuit. The RTL description for the digital popcounter is developed in Verilog and synthesized in 28 nm technology using a commercial logic synthesis tool [42]. The combined area for the FeFET-based digital LiM XOR array and the popcounter circuit is 21157 μm^2, with the XOR array and the popcounter circuit occupying 8247 μm^2 and 12910 μm^2 respectively. This design achieves a latency of 0.636 ns and consumes 20 pJ of energy to compute the final HD for 4096-bit vectors.

On the other hand, the size of the analog CAM array is constrained by the resolution of the ADC and the variability of the FeFETs [9]. In this design, the ML of each CAM array is connected to the supply (VDD) through a resistor, with the voltage drop across the resistor modulated by the number of conducting CAM cells (Fig. 14). A CAM cell conducts only when the input query bit does not match the stored bit. The number of mismatches (conducting cells) affects the ML voltage, which is then sensed by the ADC and converted into a digital output. However, as the size of the CAM array increases, so does the number of ML voltage levels the ADC must differentiate. Beyond a certain array size, the ADC can no longer distinguish between the voltage levels, limiting the scalability of the analog CAM approach. We implemented an ADC (see Fig. 14 and Fig. 18) similar to [43] with a sense margin of 20 mV which is deduced from variability and post-parasitics simulations. Therefore, if the ML voltage difference between different mismatch cases is below 20 mV, the ADC cannot accurately distinguish between the voltage levels.

SPICE simulations indicate that a maximum of 8 analog CAM cells can be accommodated without compromising the ADC sense margin. However, the variability of the FeFETs further complicates ADC sensing. In this technology, the FeFET variability exhibits a standard deviation of 57 mV in the V_{TH} distribution (Fig. 7). To evaluate the impact of variability

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT CAM ARCHITECTURES WITH THIS WORK

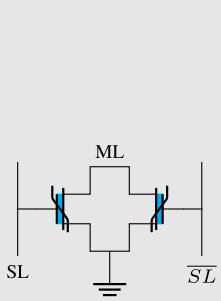
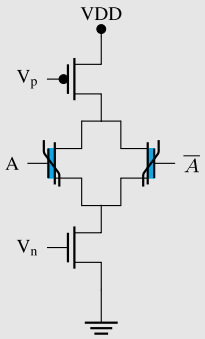
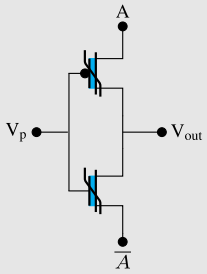
	Analog CAM	State of the art LiM XOR	This work
Cell Topology			
Non-volatility	Yes	Yes	Yes
Sensing circuit	Yes	No	No
Search voltage (V)	0.8	0.8	0.8
Total area (mm^2)	0.28 (14x)	0.04 (2x)	0.02 (1x)
Search latency (ns)	1.96 (3.3x)	0.65 (1.1x)	0.6 (1x)
Search energy (pJ)	468 (23.4x)	25 (1.25x)	20 (1x)
Write voltage (V)	± 4	± 4	± 4
Throughput (Gsamples/s)	0.51 (1x)	1.54 (3x)	1.57 (3.1x)
Efficiency (Gsamples/(s·W))	2.14 (1x)	40.4 (18x)	49.5 (23x)
Efficiency per unit area (Gsamples/(s·W·mm^2))	7.75 (1x)	972.8 (126x)	2337 (302x)

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT CAM ARCHITECTURES FOR SMALLER (1024-BIT) CLASS VECTORS

	Analog CAM	State of the art LiM XOR	This work
Total area (mm^2)	0.08 (16x)	0.006 (1.3x)	0.005 (1x)
Search latency (ns)	1.48 (2.9x)	0.52 (1.04x)	0.5 (1x)
Search energy (pJ)	88.3 (22.5x)	4.85 (1.24x)	3.92 (1x)
Efficiency (Gsamples/(s·W))	9.1 (1x)	206 (22x)	255 (28x)
Efficiency per unit area (Gsamples/(s·W·mm^2))	131 (1x)	30805 (235x)	45011 (343x)

in the CAM array, we performed Monte Carlo simulations of an 8-cell CAM array using the same variation statistics for 200 samples. As shown in Fig. 15 (a), with 8 CAM cells, increasing mismatches cause the ML voltage levels to converge and overlap due to FeFET variability, making it impossible for the ADC to distinguish between different mismatch cases accurately. Therefore, it is not feasible to implement an 8-cell CAM array with the given device-to-device variation in this technology. The maximum workable array size, with minimal ML voltage overlap, is 6-cell CAM array, as illustrated in Fig. 15 (a). Consequently, we considered a CAM array of 6 cells for the analog CAM cell-based HD calculation, as shown in Fig. 14. The ML voltage levels corresponding to the number of mismatches (or conducting CAM cells) in a CAM array of size 6 are shown in Fig. 15 (b). To calculate the HD of 4096-bit vectors, 683 6-cell CAM arrays are required, with each array connected to an ADC. The area of each 6-cell CAM array is 26 μm^2, and each 6-bit ADC occupies 372 μm^2 (Fig. 19). The ADC outputs are fed into a 2048-bit popcounter to compute the final HD. The total silicon footprint of the analog HD calculator, including the digital popcounter circuit, is 0.28 mm^2, which is 14 times larger than the proposed design. The total latency and energy dissipation of the analog design are 1.96 ns and 468 pJ, respectively. This demonstrates that the proposed digital design is nearly 3x faster and 23x more energy-efficient than the conventional analog design. A detailed comparison of the performance metrics of our design relative to the analog CAM [10] is summarized in Table II.

The proposed design achieves a throughput of 1.57 Gsamples/s, compared to 0.51 Gsamples/s for the analog counterpart, representing a 3x improvement. Furthermore, the efficiency per unit area has increased dramatically from 7.75 Gsamples/(s·W·mm^2) to 2337 Gsamples/(s·W·mm^2), marking a 302x enhancement. We also evaluated the system-level performance of the digital HD calculator by substituting our proposed digital XOR cell with the LiM-based 6T digital XOR (comprising 2 FeFETs and 4 CMOS transistors, including inverters) proposed in [33]. The latter design exhibits

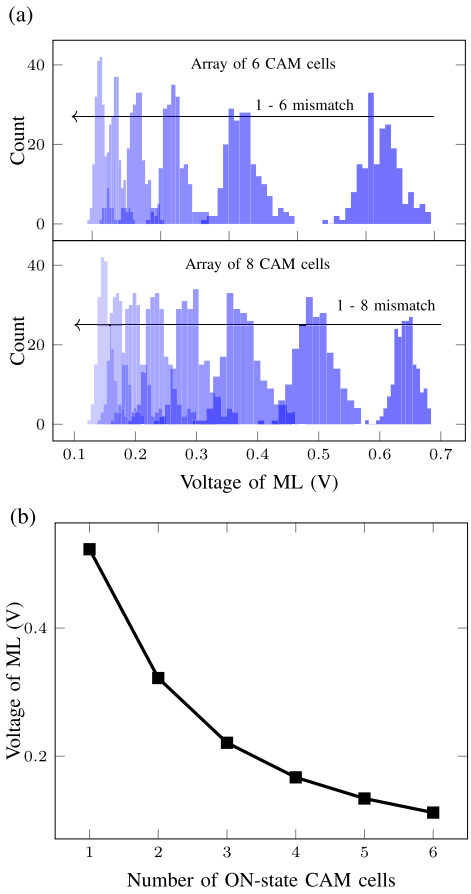


Fig. 15. Impact of variability on the design of the analog CAM circuit. (a) Variations in ML voltage for CAM arrays with 8 and 6 cells. In the 8-cell configuration, FeFET variability causes overlapping ML voltage levels during 8 mismatches, complicating ADC sensing. In contrast, the 6-cell configuration shows minimal overlap, enabling reliable ADC operation. (b) ML voltage variation with the number of conducting CAM cells (mismatches) for an array comprising 6 CAM cells.

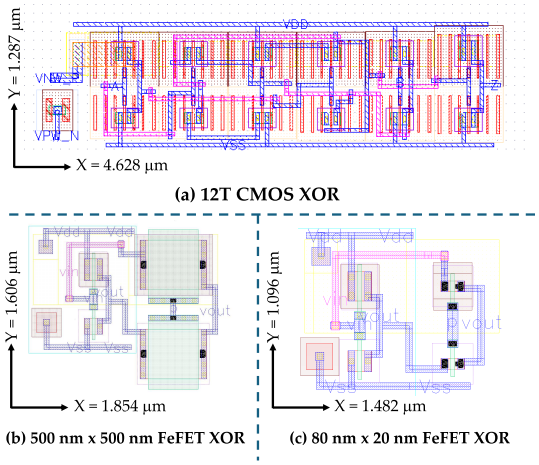


Fig. 16. Layout of XOR cells. CMOS-based XOR and proposed FeFET-based LiM XOR cell layouts designed for area comparison using the GlobalFoundries' 28 nm technology PDK. The proposed design shows significant reduction in silicon area even with the largest possible FeFET size.

an efficiency per unit area of 972.8 Gsamples/(s·W·mm^2) (Table II). In comparison, our design demonstrates a 2.4x improvement in efficiency per compute area. We have

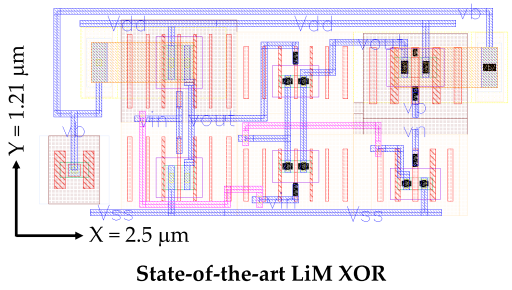


Fig. 17. Layout of state-of-the-art LiM XOR cell [33]. The layout of the FeFET-based digital XOR cell, as proposed in [33], is designed using GlobalFoundries' 28 nm technology, employing minimum feature size transistors. The design occupies an area of 3.02 μm^2, which is approximately 1.9 times larger than the area of our proposed design.

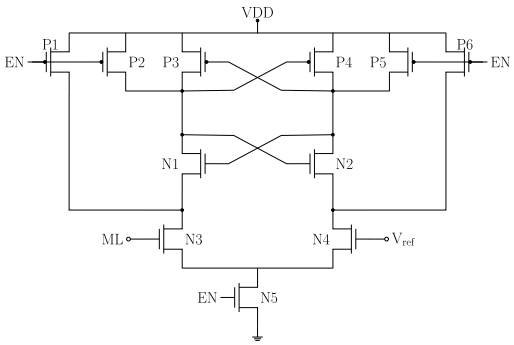


Fig. 18. Schematic of the StrongARM comparator. The comparator acts as the core of the ADC. The comparator senses the ML voltage and compares it with the V_{ref}, which is usually generated using a resistive ladder [43].

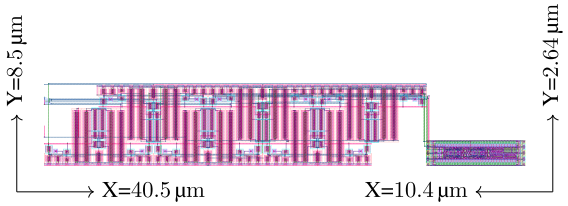


Fig. 19. Layout of the StrongARM latch comparator-based ADC. The layout of the ADC is designed using commercial EDA tool [37] with GlobalFoundries' 28 nm technology. The design occupies an area of 371.7 μm^2, which constitutes the major area of the analog CAM. The left block is the comparator stage and the right block is the thermometer to binary converter.

additionally conducted a performance evaluation for reduced array dimensions, focusing specifically on a class vector size of 1024 bits. Our analysis indicates that the proposed design demonstrates improved efficiency at smaller array sizes compared to larger ones. This improvement can be attributed to the reduced influence of peripheral circuit energy dissipation, which becomes more pronounced in larger arrays and tends to diminish the energy efficiency benefits offered by the proposed LiM cell. For instance, when the algorithm is implemented with a 1024-bit class vector, the design achieves a compute efficiency of 45,011 Gsamples/(s·W·mm^2. This corresponds to an improvement of approximately 342× over conventional analog CAM and 1.46× compared to the current state-of-the-art LiM-based design. A detailed comparison of performance metrics for all three architectures under the 1024-bit configuration is provided in Table III for reference.

V. Conclusion

This study presents a fully digital HD calculation approach utilizing a complementary FeFET-based digital LiM XOR gate. Our study accounts for FeFET device variability and provides a comparative analysis against existing analog CAM-based HD calculation circuits. The proposed design eliminates the need for sensing circuits typically required in CAM arrays, which often limit the scalability of systems. Our architecture demonstrates significant improvements in throughput and efficiency compared to conventional methodologies. Applied to HD calculations for 4096-bit class vectors, our design achieves a 23-fold reduction in energy consumption, a 3-fold decrease in latency, and a 14-fold reduction in silicon footprint compared to state-of-the-art solutions. Crucially, our FeFET-based architecture demonstrates an unprecedented efficiency of 2337 Gsamples/(s·W·mm^2), a 300-fold improvement over conventional designs, offering a unique competitive advantage where energy efficiency, reliability, and performance trade-offs have long been a concern. These findings highlight the potential of complementary FeFET-based digital LiM XOR gates in enabling efficient HD computation for HDC and other few-shot learning algorithms. The inherent resilience of our design to device variability facilitates the scaling of class vectors, which enhances the inference accuracy of HD computations.

Appendix
Design of the ADC

The presented 3-bit binary output ADC is composed of 2 main stages. In the first stage, the input analog signal is converted into a digital thermometer code. This stage is named the comparator stage (Fig. 18). A subsequent Wallace adder tree-based converter converts the thermometer code into a binary output. This second stage is called the thermometer to binary converter (therm2bin). This type of ADC is commonly known as Flash ADC. The first stage is composed of 6 identical StrongARM latch comparators [43]. The sizing of the transistors in each comparator is identical. The transistors are tuned to maximize the speed, and the input kickback noise, the input offset, and the area are minimized. The latch operates dynamically on the rising edge of the enable signal from its forced state, imposed by the precharge transistors, into a steady state where the 2 outputs of the latch have inverted values. Depending on which input transistor (N3 or N4) conducts more current, the output node connected directly to that transistor latches to 0 (GND) while the other output to 1 (VDD). To keep the assigned value until the next rising edge of the comparator Enable signal (EN) (Fig. 18), an RS latch stage is needed at the 2 outputs of the StrongARM latch [43]. The output of the RS latch is fed subsequently into the thermometer to binary converter. One input of the 6 StrongARM latches is shared between them: the input analog signal (ML). The other input are the reference voltages (V_{ref}), which are different for each comparator. These V_{ref} set the transition voltage for the output value to switch state during comparison. To reach a thermometer code, each comparator's V_{ref} is commonly linearly separated using a resistive ladder [43]. This architecture generates 6 voltages separated by a specific voltage value known as LSB. The 6 outputs of the comparator stage are fed into a Wallace adder tree decoder, which, in this case, is only composed of 2 stages. Now in binary code, the output can be easily connected to other system parts.

Compact Model Development

The FeFET device is simulated by coupling models for two components: the ferroelectric capacitor (FeCap) and the underlying intrinsic MOSFET. The intrinsic MOSFET behavior is modeled using the 28 nm PDK, while the FeCap is characterized using the Preisach-based ferroelectric model, which accurately reflects the polarization–voltage (P–V) behavior of the ferroelectric material. This macro-level modeling approach captures the nonlinear hysteresis and polarization dynamics as a function of the applied electric field. To account for the dynamic switching behavior, an auxiliary voltage (V_{aux}) is introduced. This parameter represents the effective voltage experienced by the ferroelectric dipoles after relaxation, thereby enabling more accurate simulation of switching phenomena. This is given using (1):

$$V_{\text{aux}} = V_{\text{in}} - \tau_v \frac{\mathrm{d}}{\mathrm{d}t} V_{\text{aux}} \tag{1}$$

where, τ_v denotes the relaxation time associated with V_{aux}, while V_{in} represents the external voltage applied to the gate terminal of the FeCap. The corresponding polarization state, P_{aux}, is then computed as a function of V_{aux} using (2).

$$P_{\text{aux}} = m \cdot P_s \cdot \tanh(w(V_{\text{aux}} \pm V_c)) + P_{\text{off}} \tag{2}$$

$$w = \frac{T_{\text{fe}}}{2V_c} \cdot \ln\left(\frac{P_s + P_r}{P_s - P_r}\right) \tag{3}$$

Here, V_c represents the coercive voltage of the ferroelectric material, P_s denotes the saturation polarization, P_r is the remanent polarization, m defines the slope of the polarization curve, and P_{off} accounts for the offset polarization. The direction of polarization switching - upward (−) or downward (+) - dictates the sign of the hyperbolic tangent function used in the model. The parameters m and P_{off} are extracted based on the polarization history to accurately capture the minor loop behavior in the P–V response. The resulting polarization is equated to the gate charge of the underlying MOSFET, which in turn governs its operating region. The validation of the compact model with the measured data is shown in Fig. 6. A comprehensive description of the modeling framework can be found in our prior work [34].

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